

Luong Hai Long

Computer Vision, Machine Learning, and Network Communications Intern
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Summary

B.Eng. candidate in Electronics and Telecommunications Engineering, completing the program in September 2026. Built evidence-backed academic work in YOLO/OCR inference, PyTorch signal classification, Cisco network verification, and TCP/IP-connected FPGA/SoC systems. Seeking an internship in computer vision, machine learning, or network and telecommunications engineering.

Skills

Languages and tools: Python, C/C++, MATLAB, Git/GitHub, Jupyter, Kaggle, Linux.
Computer vision, ML, and deep learning: PyTorch, YOLO/Ultralytics, OpenCV, FastALPR, OCR, object detection, 1D CNNs, scikit-learn.
Networks and communications: TCP/IP, Cisco IOS, Packet Tracer, IPv4/IPv6, VLAN, OSPFv2, DHCP, NAT, BER, OFDM, MIMO.
Embedded systems: Cyclone V FPGA/SoC, embedded Linux, STM32/ESP32, BLE, UART, I2C, SPI.

Education

VNUHCM - University of Science Sep 2022 – Expected Sep 2026
Bachelor of Engineering in Electronics and Telecommunications Engineering
Ho Chi Minh City
High-Quality Program, English-taught.
• Selected results: Introduction to Artificial Intelligence (9.1/10), Integrated Circuit Design Laboratory (9.6/10), Embedded Systems Practice (10/10), Sensors and Measurement Laboratory (9.0/10), Computer Interfacing and Data Acquisition (9.0/10), and Wireless Communications (8.5/10).

Experience

BTP Holdings Jan 2026 – Apr 2026
R&D Embedded Software Engineer Intern
Ho Chi Minh City
• Supported STM32/ESP32 firmware and board validation across C/C++ HAL modules, RTOS-oriented tasks, UART/Serial diagnostics, TCP/IP, and BLE communication paths.
• Built a C/Win32 serial diagnostic tool for board bring-up; documented USB-C PD behavior tests and power-board schematic observations to support firmware-hardware debugging.

Robotics & IoT Club, VNUHCM - University of Science Nov 2023 – Jul 2024
Technical Team Member
Ho Chi Minh City
• Assembled and tested robotics and IoT prototypes; isolated sensor wiring, power, and firmware issues during bring-up with C/Lua test routines.

Anh Trung Smartphone Store / Xuan Anh Computer 2023 – 2025
Repair Technician (Part-time)
Ho Chi Minh City
• Diagnosed phone, laptop, and PC faults across power, display, storage, thermal, connectors, software, and peripherals; completed functional checks before handover.

Selected Projects

Vietnamese Automatic License Plate Recognition with YOLO and OCR Jan 2026 – Apr 2026
Team of seven | Python, PyTorch, YOLO, OpenCV, FastALPR, OCR, FFmpeg, Tkinter
• Co-developed Python desktop image/video inference and a bounded PlateGate LAN whitelist demo; maintained Kaggle/IPYNB training evidence, Git LFS assets, and release packaging.
• The YOLO detector reached 99.45% mAP at 0.5 IoU, 99.45% precision, and 99.37% recall on the project validation set; the work is an academic dataset and LAN-prototype demonstration.

Automatic Modulation Classification with DSP Features and 1D CNN Aug 2026
Python, PyTorch, scikit-learn, Random Forest, Streamlit, I/Q signal processing
• Built a six-class I/Q signal-classification workflow with 18 DSP features, a 400-tree Random Forest, and an 81,030-parameter 1D CNN.
• Validation-selected probability fusion reached 73.48% accuracy and 73.40% macro-F1 on a fixed 1,320-frame synthetic-data test split.

Network Technology Laboratory 2026
Cisco IOS, Packet Tracer, IPv4/IPv6, VLAN, STP, EtherChannel, OSPF, DHCP, NAT, WLAN
• Configured and verified six laboratory networks through routing tables, VLAN/trunk state, OSPF adjacencies, DHCP addressing, NAT translation, wireless access, and end-to-end reachability.
• Produced reproducible Typst reports and a PowerShell verification workflow for the required artifacts.

SoC Ethernet Controller on Intel DE10-Standard Jan 2026 – Apr 2026
Team of five | Cyclone V, HPS/Linux, VHDL, TCP/IP, Ethernet, Python/Tkinter
• Co-developed a LAN prototype in which PC/Android clients send short TCP commands to an HPS Linux daemon, which writes through the lightweight HPS-to-FPGA bridge to drive HEX0-HEX5 displays.
• Scope: course prototype only; no internet/NAT traversal, custom kernel driver, or production security hardening.